PIERRE MAKHLOUTA

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EDUCATION

York University, Lassonde School of Engineering Sep 2024 - May 2028

Specialized Honours BSc, Computer Science

Coursework: Data Structures, Advanced OOP, Software Tools, Computer Organization, Algorithms (in progress)

TECHNICAL SKILLS

Cloud and DevOps: AWS (EC2, S3, IAM, Systems Manager, RDS, ALB), Terraform, Docker, Kubernetes, GitHub Actions, nginx, Grafana
Languages: Python, TypeScript, Bash, SQL, PL/pgSQL, PowerShell, C, Java
Systems: Linux, macOS launchd, Tailscale
Backend and Data: PostgreSQL, SQLite, Supabase, FastAPI, pandas, ETL, star schema, Git, Excel (Power Query), Power BI
Testing: pytest, Vitest, pgTAP, Playwright, axe-core
Security: TLS, OIDC, Row Level Security, Web Crypto API, Argon2id, bcrypt, CSRF protection, secret scanning, least-privilege IAM

PROJECTS

PSells: Inventory, Listing and Analytics Platform on AWS github.com/pierremakhlouta/psells

Python, FastAPI, PostgreSQL, pandas, Docker, nginx, Terraform, GitHub Actions, AWS, Kubernetes, Grafana

- Built and run the software for my own reselling business, used daily by two people: a web app, terminal app and HTTP API on PostgreSQL that share one set of business rules, with Argon2id login, CSRF tokens and nginx with TLS 1.3
- Built a pandas ETL that rebuilds a star-schema warehouse every hour through a read-only role, committing only if its totals match the live dashboard; SQL views over it feed an analytics page
- Defined all 23 AWS resources in Terraform with locked, versioned S3 state; no SSH (Session Manager only), secrets in Parameter Store and a least-privilege IAM role. Built RDS and an ALB, proved them, then switched them off to stay within free credits
- Wrote 1,600+ tests against a real PostgreSQL; six GitHub Actions workflows lint, scan (gitleaks, Grype), test on a throwaway Kubernetes cluster and deploy by digest through OIDC with no stored AWS keys
- Set up Grafana Cloud monitoring in Terraform with checks from 3 regions and availability and latency SLOs; ran a planned 15-minute database outage, the alert fired in about 5 minutes, and wrote a blameless postmortem
- Automated daily backups that are restore-tested before being kept and copied to an encrypted S3 bucket the server can write to but not read; launchd jobs also rebuild analytics and warn before certificates expire

Evenly: Shared Expense Tracker github.com/pierremakhlouta/evenly

TypeScript, Vite, PWA, Supabase (PostgreSQL, PL/pgSQL), Deno Edge Functions, GitHub Actions, Vitest, pgTAP, Playwright

- Built a solo installable web app, used daily, for splitting bills between the two households in my house
- Put the backend in about 28 PL/pgSQL functions; the browser cannot read any table directly (Row Level Security with zero policies, an allowlist of functions), and a CI test fails if anything else becomes callable
- Implemented Web Push from scratch with the Web Crypto API (VAPID, RFC 8291/8292), verified by interop tests; kept all money in integer cents, with the same math in TypeScript and SQL agreeing to the cent
- Wrote about 670 tests, including Playwright runs on emulated iPhone and Android with WCAG 2.2 AA checks

EXPERIENCE

Data Analyst, MES Support (Co-op) | S&C Electric, Toronto Aug 2026 - Present

- Built a live Excel tracker of every job on the shop floor, now used by more than 50 people
- Designed a shipping skid sizing calculator used by the whole shipping team; it makes sizing 25% faster
- Built a Crating Tracker that uses Power Query to auto-fill sales order and customer from the job number
- Designed a yearly performance-review tracker for supervisors across 4 review categories, with a summary dashboard
- Gathered network drop requirements for a new paint line building from layout drawings, for next year's budget

Inventory and Procurement Summer Student | S&C Electric, Toronto May 2026 - Aug 2026

- Tracked stock and coordinated parts ordering across teams for high-voltage power distribution production

Production and Logistics Operations Summer Student | S&C Electric, Toronto May 2025 - Aug 2025

LEADERSHIP

Sponsorship Director | Syrian Students Association, York University Sep 2026 - Present

- Find and secure sponsors for the association's events and activities

CERTIFICATES AND TRAINING

- Microsoft 365 and Agent Administration (AB-900) Specialization (Microsoft on Coursera, Oct 2026)
- Google IT Automation with Python Professional Certificate (Google, Sep 2026)
- Learning Linux for LFCA Certification Specialization (LearnQuest, Jun 2026)